

Vincent Liu

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

Masters of Science in Computer Science, Specialization in Machine Learning — GPA: 4.0 Jan. 2025 – Dec. 2026

- Courses: Machine Learning, Reinforcement Learning, Computer Vision, Machine Learning for Trade, Robotics: AI Techniques, Graduate Algorithms, Quantum Computing, Game AI, Video Game Design

University of Georgia

Athens, GA

Bachelors of Science in Computer Science — GPA: 3.94 Aug. 2021 – Dec. 2024

- Courses: Data Mining, Artificial Intelligence, Evolutionary Computation, Data Structures, Software Engineering

EXPERIENCE

Data Science Intern

Feb 2025 – Dec 2025

N&C Wholesale

Duluth, GA

- Analyzed customer purchasing and inventory data using **Python, Pandas, NumPy, SQL, and Jupyter**, performing data cleaning, feature engineering, and EDA to identify demand trends and customer preferences
- Built and evaluated **machine learning models with Scikit-learn, TensorFlow, and PyTorch** for demand forecasting and customer behavior prediction, improving inventory optimization and reducing stock issues
- Developed dashboards and automated reports using **SQL, Tableau, Power BI, and Excel**, visualizing sales performance, customer segmentation, and inventory KPIs to support business decisions

Undergraduate Research Assistant

May 2023 – March 2025

University of Georgia

Athens, GA

- Synthesized findings from **150+ ITS/RTS studies**, analyzing **AI, LLM, ML, and HCI methodologies** used to deliver personalized, real-time student feedback and adaptive learning support
- Designed and executed ITS/RTS evaluation experiments, tracking **200+ learner interactions**, measuring engagement, retention, and learning outcomes, with retention improving by **15–20% above baseline**
- Engineered and fine-tuned an **LLM-based physics tutoring system**, leveraging **prompt engineering, adaptive feedback generation, and model optimization** to improve K–12 student problem-solving accuracy by **18% in pilot trials**

Software Engineer

August 2020 – May 2021

Oracle

Cumming, GA

- Developed and deployed a **full-stack Python application** to automate monthly electrical and water bill calculations, reducing manual processing time by **50%** while improving billing accuracy and operational efficiency
- Architected back-end workflows using **Python, SQL, HTML, and CSS**, building data pipelines, database integrations, and interactive front-end visualizations for 100+ users to analyze billing and usage trends

PROJECTS

AWS DeepRacer

March 2026 – May 2026

- Designed and trained autonomous racing agents using **Deep Reinforcement Learning, PyTorch, and AWS DeepRacer Gym** across Time-Trial, Object Avoidance, and Head-to-Head environments
- Engineered custom **reward functions, CNN-based state representations, and hyperparameter tuning** using stereo camera and LiDAR data to improve lap completion and policy robustness

Overcooked Multi-Agent RL Project

Feb 2026 – March 2026

- Developed cooperative **multi-agent reinforcement learning** agents in **PyTorch** for the Overcooked environment, optimizing teamwork across multiple kitchen layouts
- Designed custom **reward shaping, policy updates, and training analysis** to improve coordination, credit assignment, and overall agent performance across all layouts

TECHNICAL SKILLS

Machine Learning: Python, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, SQL, Reinforcement Learning, Computer Vision, NLP, LLMs

Software Engineering: Java, C#, AWS, React.js, JavaScript, HTML/CSS, MongoDB, MySQL, Git, Node.js

Game Development: Unity, Unreal Engine, Blender, VR/AR